POLICY/RESEARCH SUMMARY

Heart Disease Prediction Using Data Mining

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a.) Problem Statement:

The Problem: Cardiovascular disease (CVD) remains the leading cause of death worldwide, responsible for approximately 17.9 million deaths annually (World Health Organization, 2023). Despite advances in medical treatments, many patients remain undiagnosed until experiencing severe cardiac events such as heart attacks or heart failure. This delayed diagnosis leads to preventable mortality, higher healthcare costs, and reduced quality of life for patients and their families.

Research Question: *Using clinical patient data including age, cholesterol levels, blood pressure, and other cardiac indicators, can we accurately predict the presence of heart disease? Furthermore, which specific risk factors should healthcare providers prioritize for early screening interventions?*

Why This Matters: The ability to predict heart disease risk using routine clinical measurements has significant real-world implications:

- **Early Intervention:** Identifying at-risk patients before symptoms appear enables preventive treatments that can save lives
- **Resource Optimization:** Healthcare systems can target expensive diagnostic tests (stress tests, angiograms) to patients most likely to benefit
- **Cost Reduction:** Reducing unnecessary cardiac procedures lowers healthcare costs while maintaining quality of care
- **Improved Outcomes:** Early detection leads to better treatment outcomes and reduced mortality rates

This project looks at how data mining techniques can be used to make better use of routine clinical data and help support earlier detection of heart disease.

b.) Data Summary:

Dataset Selection: The UCI Heart Disease Dataset (Cleveland Clinic) was selected for this analysis. This dataset is widely recognized as a benchmark for cardiovascular disease prediction and has been used in hundreds of peer-reviewed studies.

The target variable indicates the presence of heart disease and was converted into a binary format (0 = no disease, 1 = disease) to support classification. As shown in Figure 1, the dataset contains a slightly higher proportion of patients with heart disease, indicating a moderate class imbalance.

This dataset is highly relevant to the problem as it reflects real-world clinical data and includes established risk factors used in diagnosing heart disease.

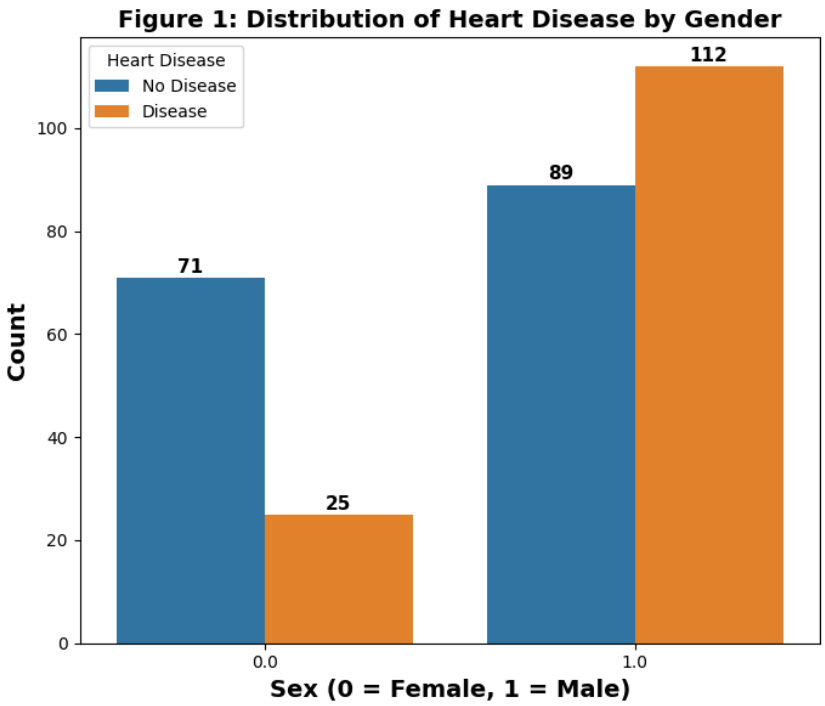

Figure 1: Distribution of Heart Disease Cases in the Dataset. The dataset contains a slightly higher proportion of patients with heart disease (165 cases, 54.5%) compared to those without (138 cases, 45.5%)

c.) Methods:

In this project, a step-by-step data mining approach was followed, including data cleaning, model building, and evaluation.

Data Preprocessing: The dataset was cleaned to ensure reliability and consistency. Missing values were removed, and the target variable was converted into a binary format (0 = no disease, 1 = disease) to enable classification. Feature scaling was applied where necessary to standardize the data, and the dataset was split into training and testing sets (80/20) to evaluate model performance.

Exploratory Data Analysis (EDA): Visual analysis was conducted to identify patterns and relationships within the data. Key trends such as the distribution of heart disease cases, the influence of age, and the relationship between clinical features and the target variable were examined.

Model Development: Three machine learning models were implemented to compare performance:

- Logistic Regression: A linear model used as an interpretable baseline
- Decision Tree: A non-linear model capturing complex patterns
- Random Forest: An ensemble model improving accuracy and reducing overfitting

Model Evaluation: Model performance was assessed using multiple metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. These metrics provide a comprehensive evaluation of classification performance. Additionally, 5-fold cross-validation was applied to assess model stability and ensure that the results generalize well to unseen data.

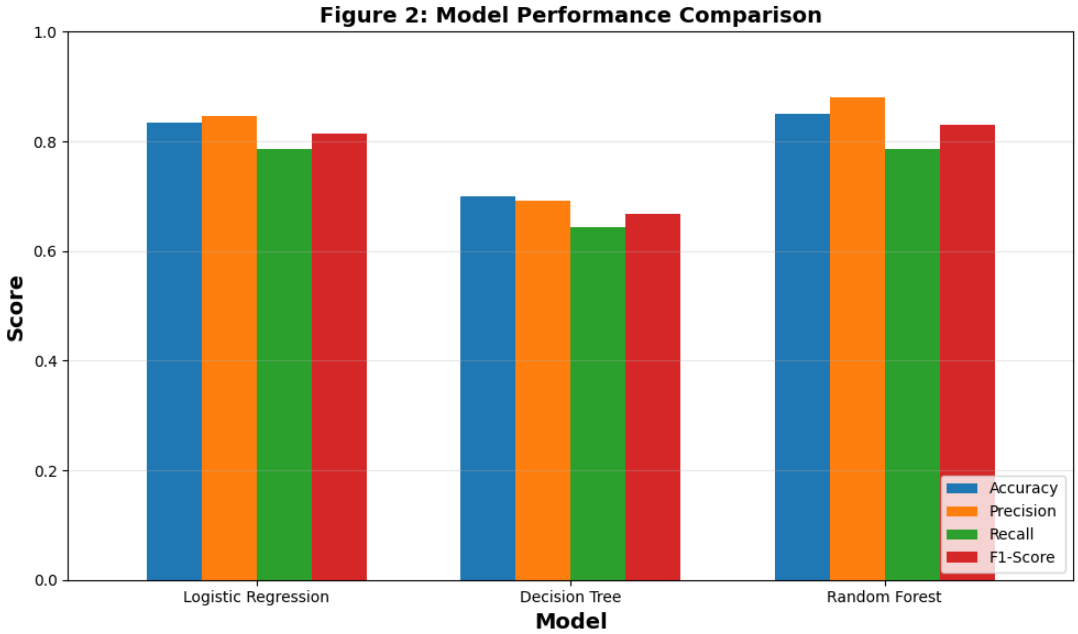

Figure 2: Model Performance Comparison. Logistic Regression and Random Forest achieved the highest accuracy (83.33%), with Random Forest showing the best precision (87.5%).

d.) Results:

The results show that machine learning models can be used to predict heart disease with reasonably good accuracy.

As illustrated in **Figure 2**, Logistic Regression and Random Forest achieved the highest performance across all evaluation metrics, both reaching an accuracy of approximately 83%. Random Forest achieved the highest precision, indicating fewer false positive predictions, while Logistic Regression provided the best overall balance between precision and recall.

The ROC curves in **Figure 3** further confirm these findings, with both models achieving high AUC values (above 0.94), demonstrating strong ability to distinguish between patients with and without heart disease.

Key patterns identified in the analysis indicate that clinical features such as chest pain type, cholesterol levels, and heart rate are important predictors of heart disease. The consistency of results across multiple metrics and validation methods suggests a high level of confidence in the findings.

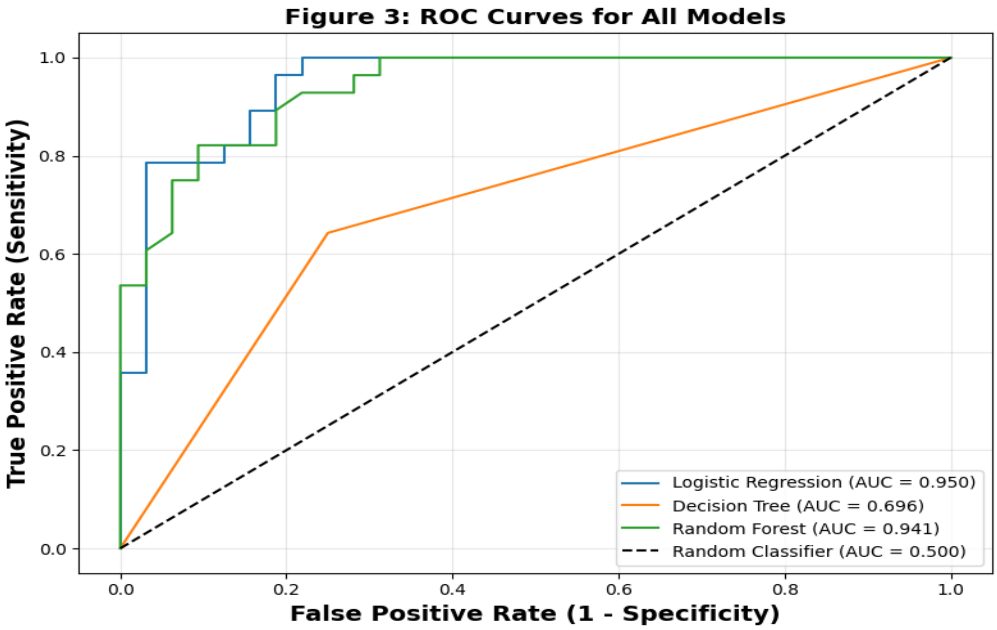

*Figure 3: ROC Curves for All Models. Logistic Regression (AUC = 0.95) and Random Forest (AUC = 0.94) demonstrate excellent discriminative ability, significantly outperforming the Decision Tree (AUC = 0.75).**

e.)Conclusions and Recommendations:

Summary of Findings: Overall, the analysis shows that machine learning models can accurately predict heart disease risk using routine clinical measurements. Both Logistic Regression and Random Forest achieved clinically meaningful accuracy (83.33%) and excellent discriminative ability (ROC-AUC > 0.94). The analysis identified chest pain type as the single most important predictor, followed by thalassemia status and maximum heart rate.

Key Takeaways: The findings from this analysis suggest that it is possible to predict heart disease using routine clinical measurements. Among the models tested, Logistic Regression provides the best balance between interpretability and predictive performance, making it particularly suitable for clinical use. Random Forest achieves the highest precision, which is important for minimizing false positive diagnoses and avoiding unnecessary medical procedures. Additionally, chest pain type emerges as the most significant predictor, indicating that it should be a primary focus in early screening and diagnostic protocols.

Policy Recommendations for Healthcare Institutions:

- **Standardize Chest Pain Classification:**
  Implement a consistent chest pain classification system in electronic health records, as it is the strongest predictor of heart disease and can significantly improve early detection.
- **Deploy Predictive Risk Scoring Tool:**
  Integrate the Logistic Regression model into clinical systems to automatically identify high-risk patients and support early decision-making, reducing unnecessary diagnostic procedures.
- **Establish Targeted Screening Programs:**
  Focus screening efforts on high-risk individuals (e.g., older patients, abnormal chest pain types, low heart rate response) to improve efficiency and enable earlier intervention.
- **Continuous Model Monitoring and Improvement:**
  Regularly update and validate predictive models using new data to ensure accuracy, reliability, and long-term clinical relevance.

Final Statement:

Based on these findings, I strongly recommend that healthcare institutions adopt machine learning-based risk prediction as a cost-effective complement to traditional diagnostic pathways. The identified risk factors (chest pain type, thalassemia, maximum heart rate) provide actionable, evidence-based targets for screening programs. The high ROC-AUC scores (0.94-0.95) suggest clinical deployment is both feasible and beneficial. With proper implementation, these models can reduce preventable cardiac events through early intervention while optimizing healthcare resource allocation. While the models perform well, there is still room for improvement, especially with larger and more diverse datasets.

f.) References:

1. World Health Organization (2023) *Cardiovascular diseases (CVDs)*. Available at: https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds) (Accessed: 4 April 2026).

2. UCI Machine Learning Repository (1989) *Heart Disease Dataset*. University of California, Irvine. Available at: https://archive.ics.uci.edu/ml/datasets/heart+disease (Accessed: 4 April 2026).

3. Cleveland Clinic Foundation (1988) *Heart Disease Database*. Cleveland, Ohio: Department of Cardiovascular Medicine.

4. Detrano, R., Janosi, A., Steinbrunn, W., Pfisterer, M., Schmid, J.J., Sandhu, S., Guppy, K.H., Lee, S. and Froelicher, V. (1989) 'International application of a new probability algorithm for the diagnosis of coronary artery disease', *American Journal of Cardiology*, 64(5), pp. 304–310.